

Experiment-13

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Course Code: -MLA0305

Slot: B

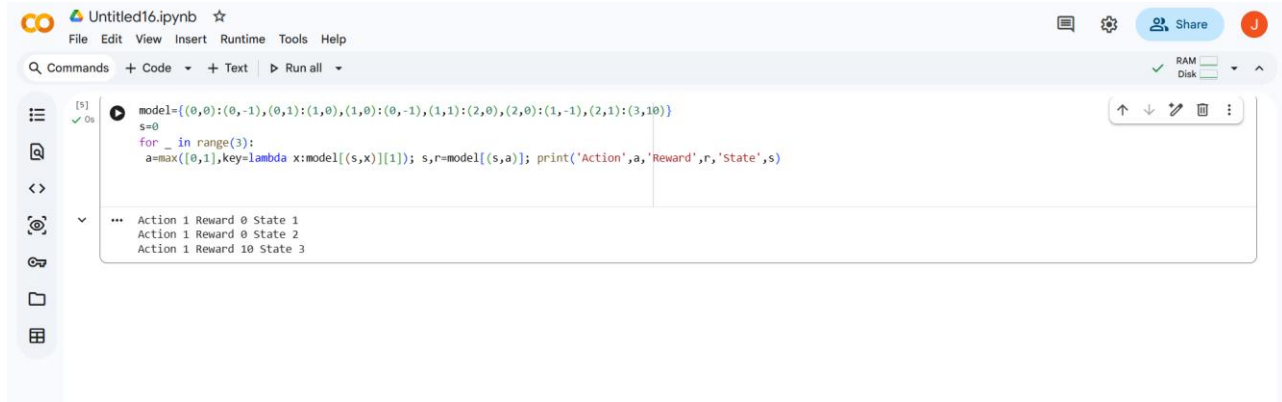
Title: Development of a Model-Based Reinforcement Learning Agent

Aim:

To develop a model-based Reinforcement Learning agent capable of learning environment dynamics and performing planning for optimal decision-making.

Procedure:

1. Define or select a suitable environment.
2. Initialize the transition and reward models.
3. Allow the agent to interact with the environment.
4. Collect state, action, reward, and next-state observations.
5. Update the learned environment model.
6. Use the learned model to simulate future states.
7. Perform planning using simulated transitions.
8. Select actions based on predicted future rewards.
9. Continue learning and updating the model.
10. Evaluate the agent's performance.



The screenshot shows a Jupyter Notebook window titled "Untitled16.ipynb". The interface includes a top menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu is a toolbar with "Commands", "Code", "Text", and "Run all" buttons. On the right side of the toolbar, there are icons for RAM and Disk usage, a "Share" button, and a user profile icon. The left sidebar contains navigation icons for the notebook's structure. The main area displays a code cell with the following Python code:

```
model={('0,0):(0,-1),(0,1):(1,0),(1,0):(0,-1),(1,1):(2,0),(2,0):(1,-1),(2,1):(3,10)}
s=0
for _ in range(3):
    a=max([0,1],key=lambda x:model[(s,x)][1]); s,r=model[(s,a)]; print('Action',a,'Reward',r,'State',s)
```

The output of the code cell is displayed below the code, showing three lines of text:

```
... Action 1 Reward 0 State 1
Action 1 Reward 0 State 2
Action 1 Reward 10 State 3
```

Result:

The agent successfully learned an approximate model of the environment and used the model for planning and decision-making.